



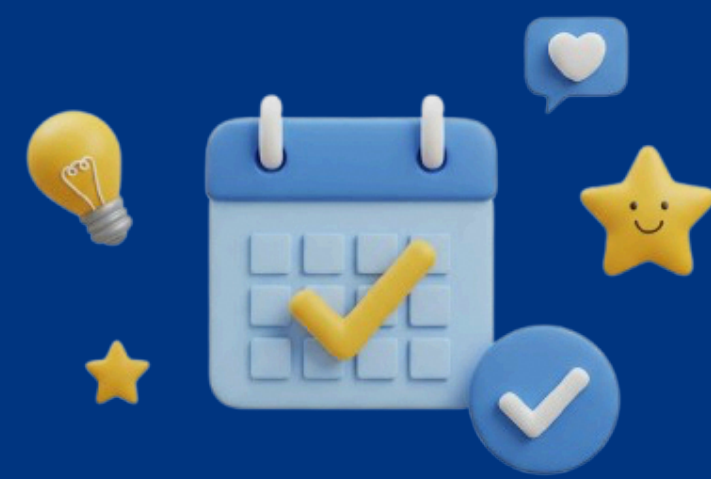
Booking.com

Business Case Study and Report

Customer Satisfaction Analysis in European Hotels
What truly influences traveler satisfaction?

Agenda

- ★ Introduction
- ★ Axis 1 — Impact of Traveler Type on Customer Satisfaction
- ★ Axis 2 — Seasonal effect on satisfaction
- ★ Axis 3 — Impact of review count on rating severity
- ★ Conclusion & Recommendations
- ★ Appendix





Introduction

Context, Objectives, and Study Methodology



[Back to Agenda](#)



Initial Findings

Dataset Overview

516,000 traveler reviews
from hotels across Europe

Source

Booking.com reviews collected via Kaggle
[Dataset link](#)

Observations

Voluntary response bias identified

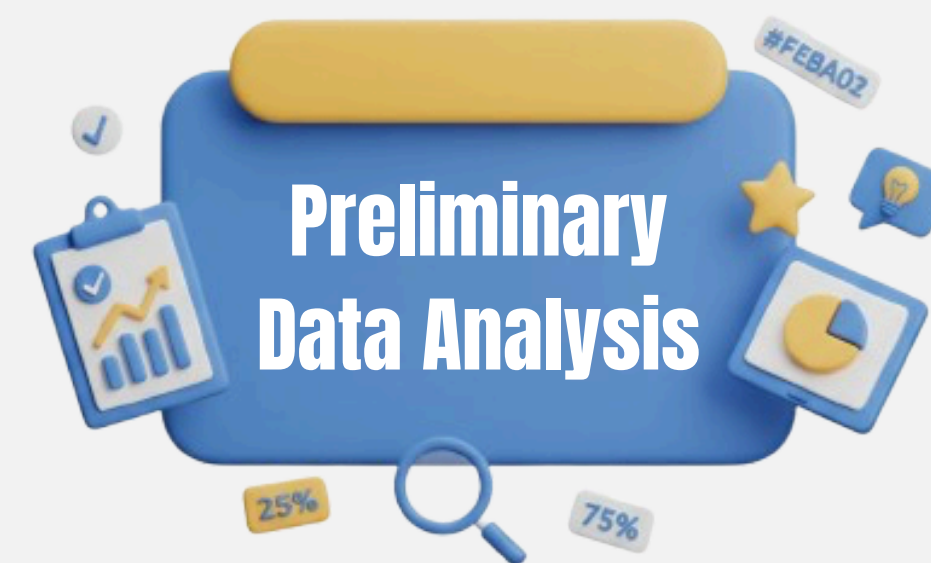
Overall high satisfaction scores 70% of
the initial dataset retained after cleaning



Challenges

Tag breakdown for content analysis

Postal code positioning in hotel
addresses (United Kingdom)



Preliminary Data Analysis

Data cleaning

No outliers detected

Columns “days since last review” and
“hotel GPS coordinates” removed



Axis 1

Impact of Traveler Type on Customer Satisfaction

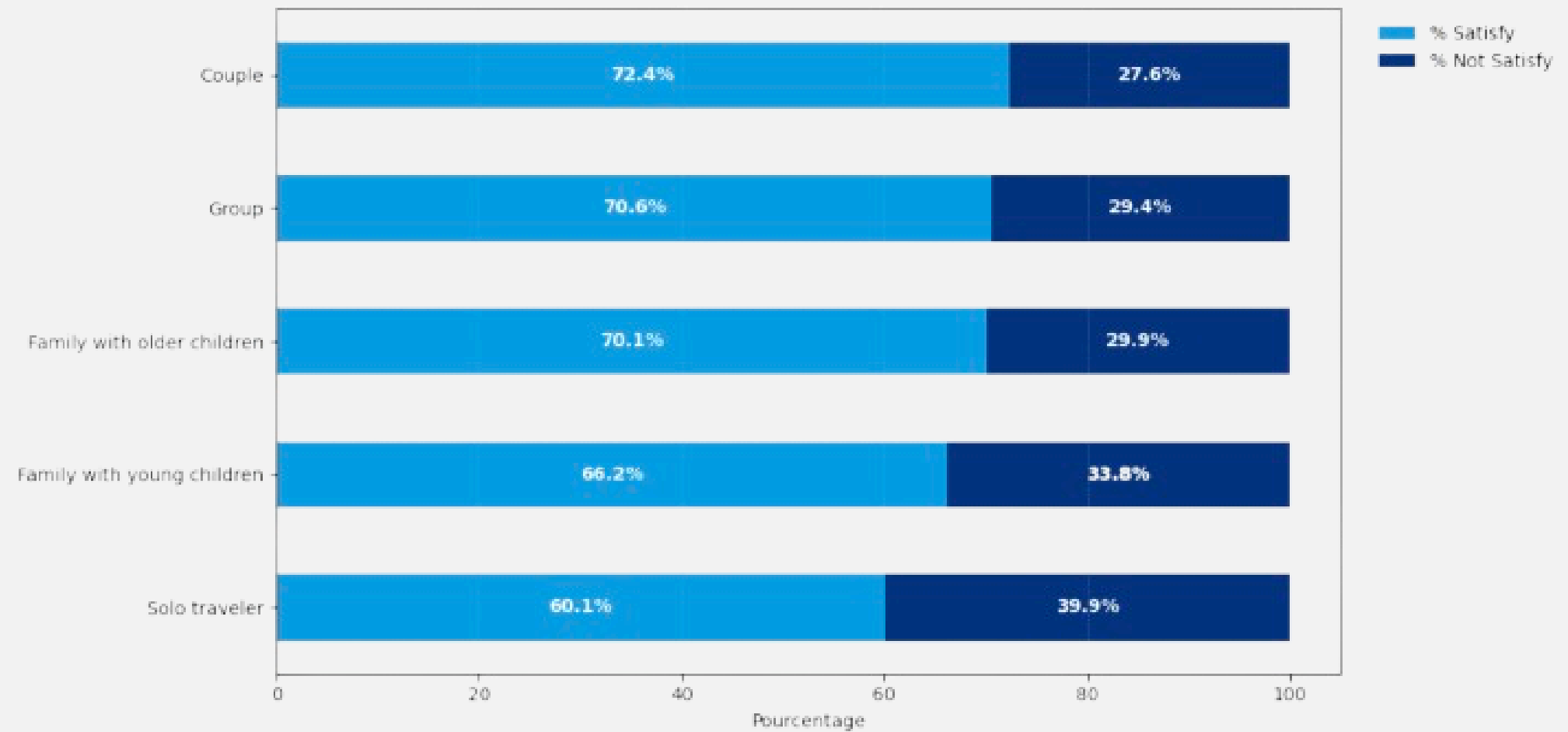
Customer profile segmentation analysis



Most satisfied traveler types



Shared trips (couples, groups, families) show higher satisfaction levels, while solo travelers report the most dissatisfaction.



Leisure vs. Business Stays

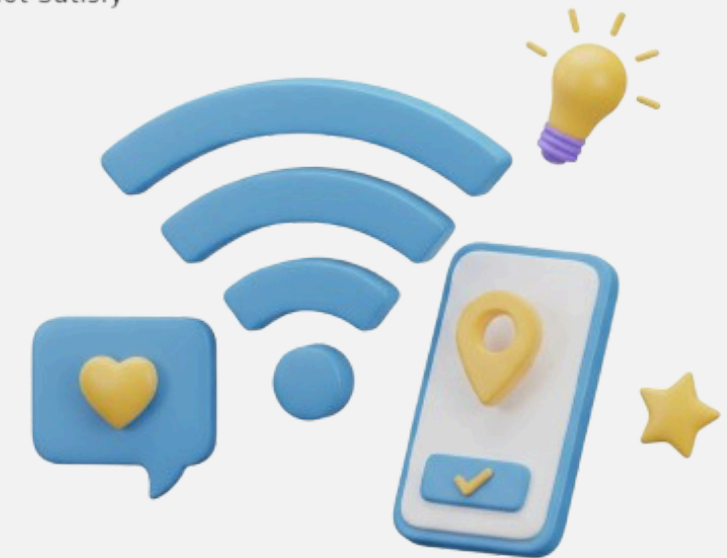
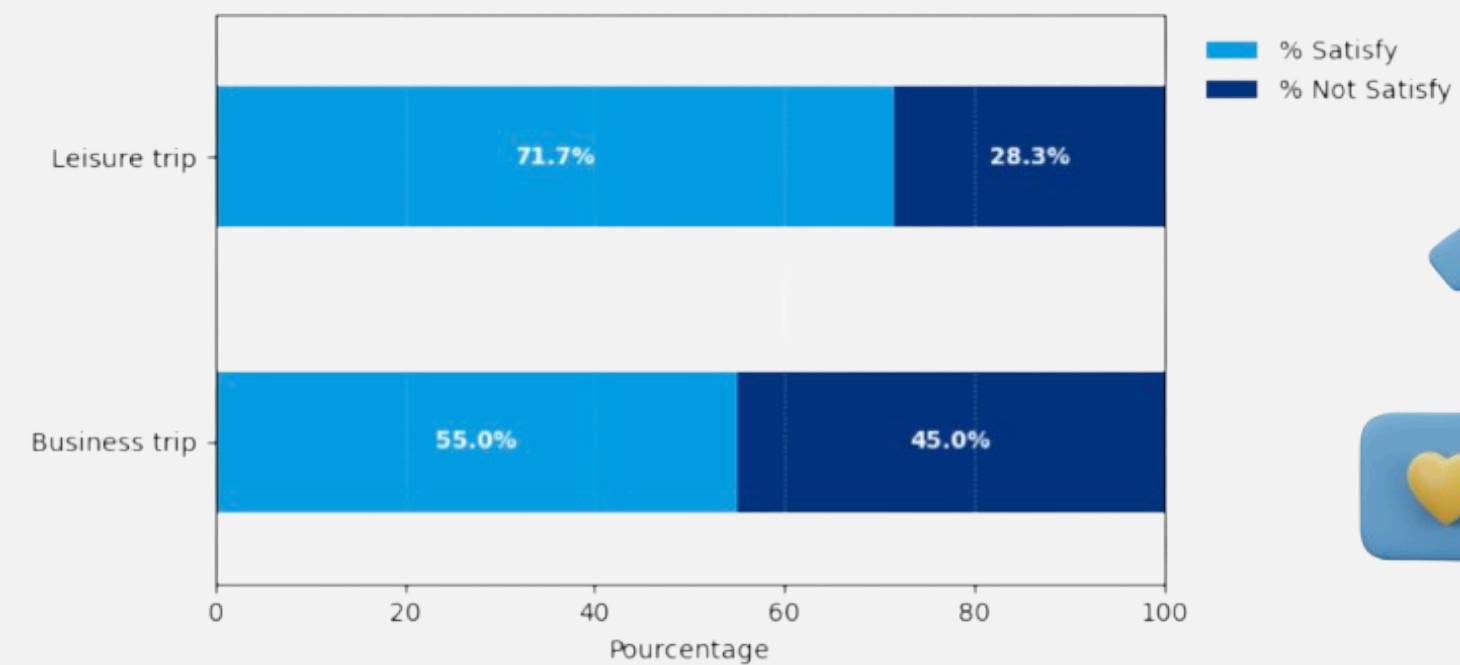


Leisure trips

High satisfaction explained by a relaxed mindset and enjoyment-focused expectations

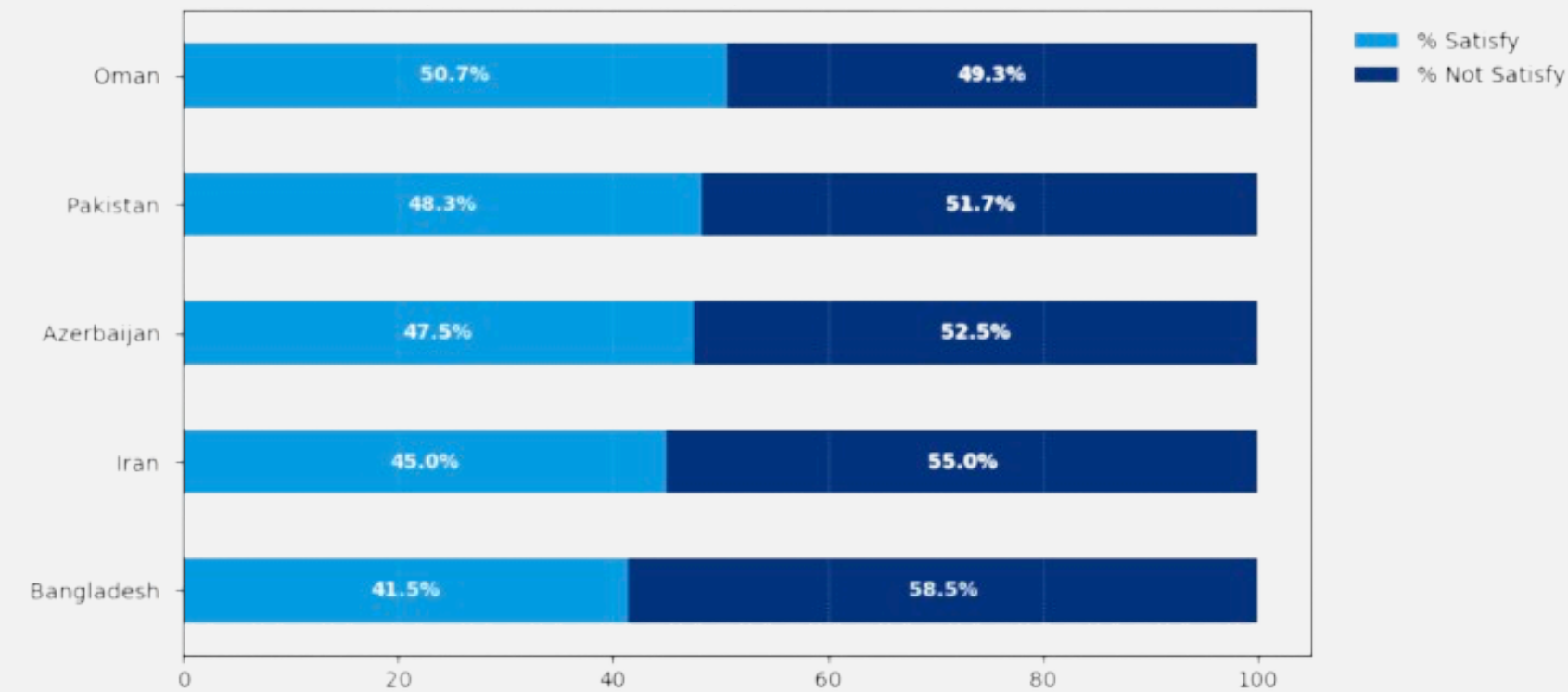
Business trips

Higher expectations regarding efficiency and professional service

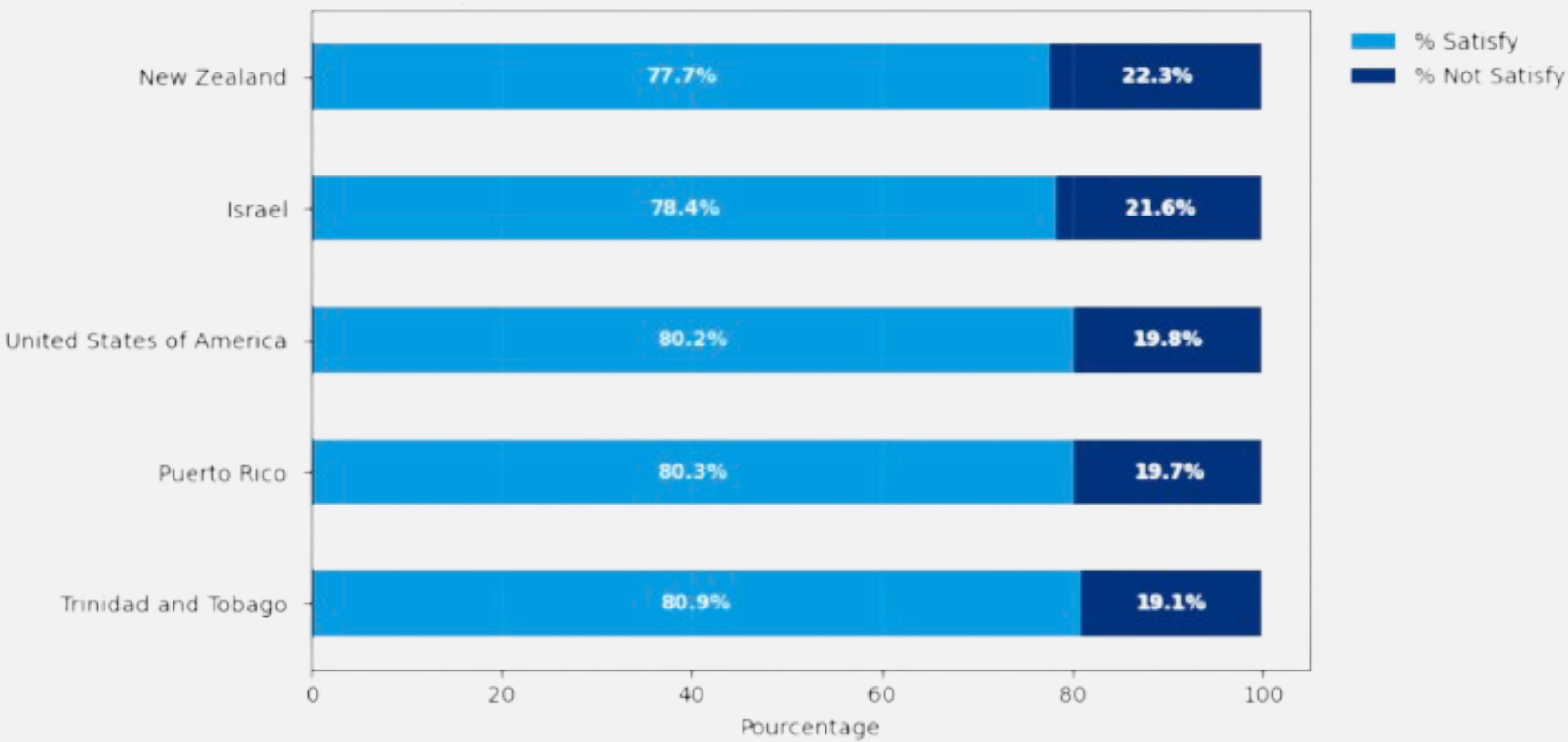




Top 5 strictest nationalities (> 50 reviews)



Top 5 most satisfied nationalities (> 50 reviews)



Cultural factors

Expectations vary across national cultures and rating habits. These results suggest that some nationalities adopt a more positive attitude toward their experience, which may reflect different expectations or a more enthusiastic rating behavior.

Impact of room type on rating



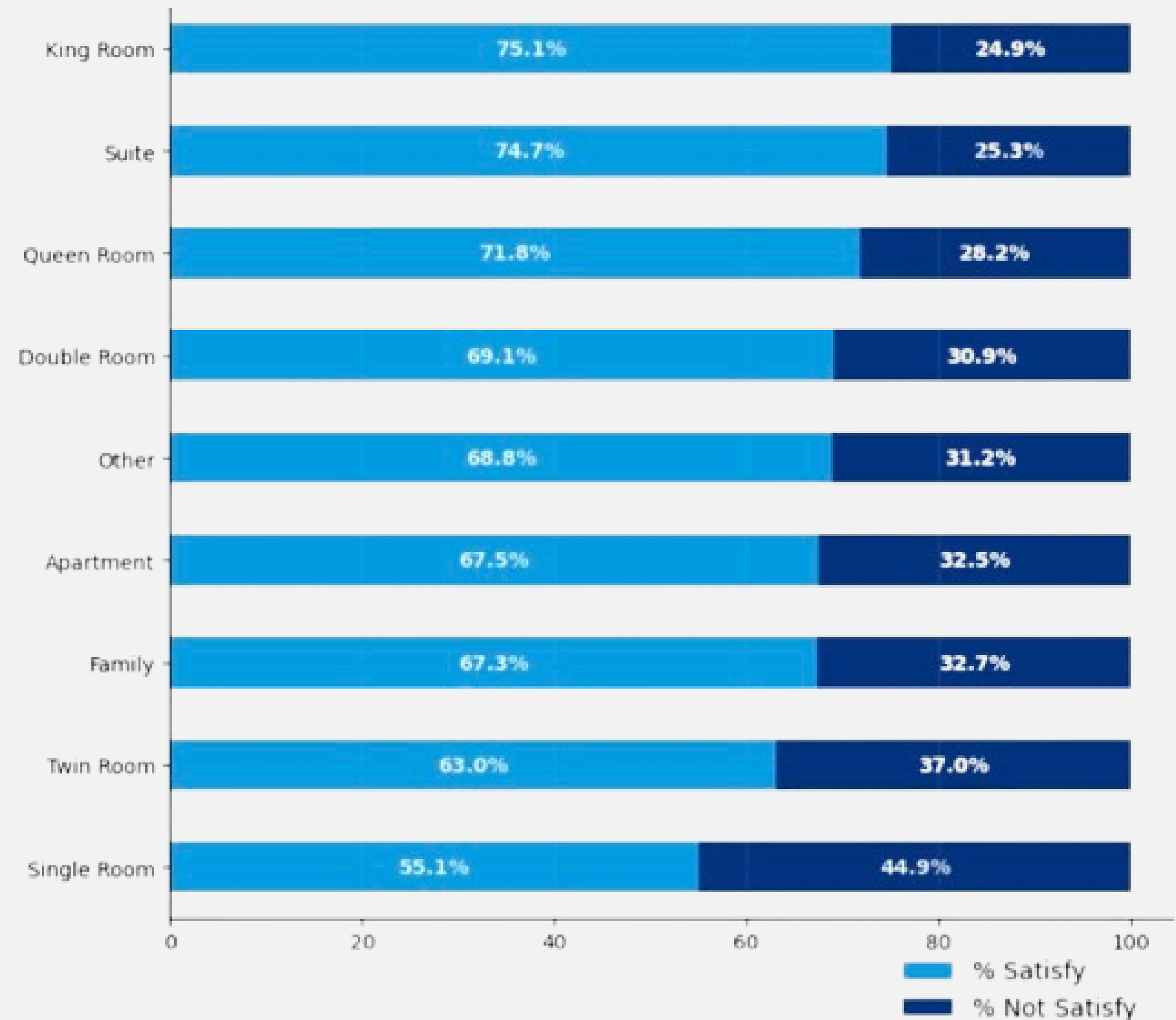
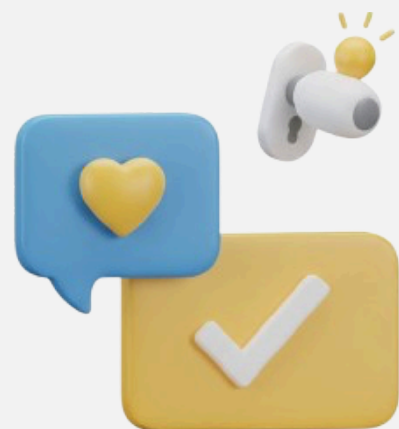
Let's compare our ranking with the ranking of room types by decreasing level of standing in Europe:

- Suite – 2nd place
- Apartment – 6th place
- King Room – 1st place
- Queen Room – 3rd place
- Family Room – 7th place

King Rooms stand out with unusually high satisfaction levels, even exceeding Suites.

Single Rooms reflect lower ratings similar to Solo Travelers.

If linked to the same hotel group, a review of Family Room and Apartment services is recommended to address weaker satisfaction.





Axis 2

Seasonal effect on satisfaction

Analysis of temporal variations and seasonal patterns

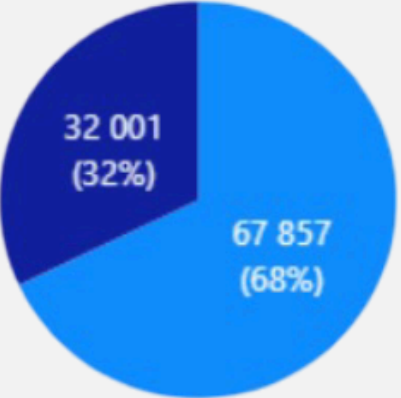




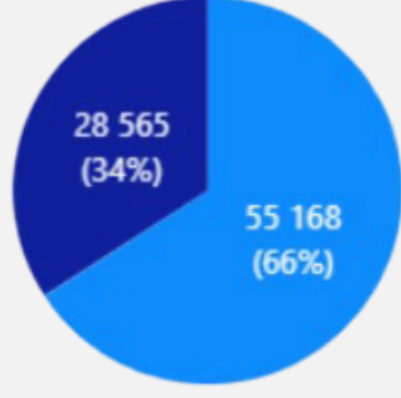
Satisfaction
by season



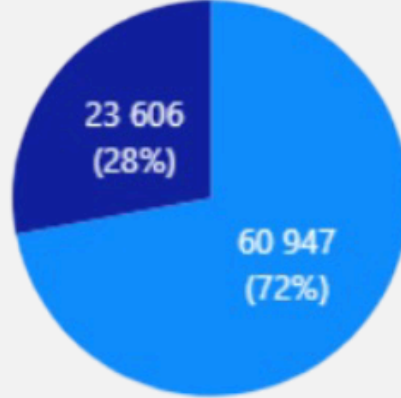
Summer



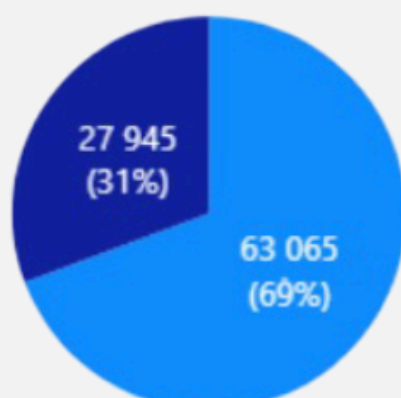
Autumn



Winter



Spring



● Satisfy
● Not satisfy

● Satisfy
● Not satisfy

● Satisfy
● Not satisfy

● Satisfy
● Not satisfy

Customer satisfaction remains stable throughout the year, with a slight peak in winter (72%) and a dip in autumn (66%). Low seasons appear to provide a better experience, likely due to lower crowd levels.





Axis 3

Impact of review count on rating severity

Observing trends between review quantity and rating intensity





Rating severity by user activity level

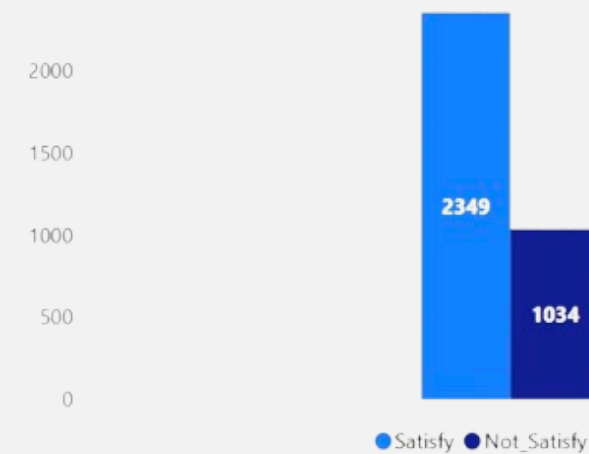


Highly Active Users

73

Average rating among 3,383 highly active users.

Mostly satisfied users — frequent reviewers tend to rate more positively.

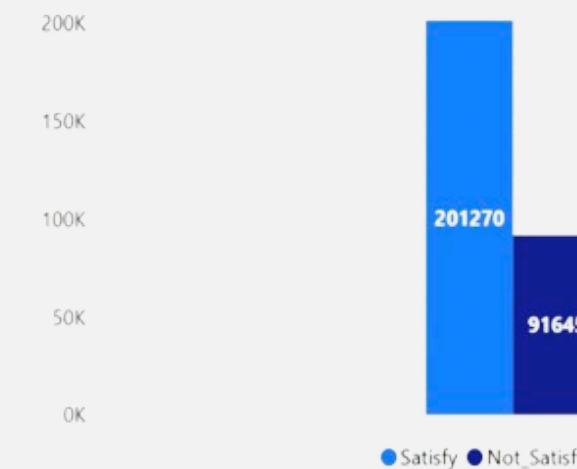


Less Active Users

3

Average rating among 292,915 less active users.

Higher proportion of dissatisfied users among infrequent reviewers



Active users give more balanced ratings, while less active users express stronger dissatisfaction.



Conclusion & Recommendations



Dataset limitations



Missing information

Key hotel and traveler details unavailable



Voluntary response bias

Extreme experiences overrepresented



Limited timeframe

No long-term trend analysis possible



Data quality variability

Missing or inconsistent values

Recommendations



Solo travelers & Business trips

Focus on solo and business travelers — satisfaction levels are **6 points** lower than couples and families.



Room type feedback

Analyze feedback by room type — King Rooms show about **75%** satisfied guests, compared to around **55%** for Single Rooms.
These gaps may help identify key levers to improve overall satisfaction.

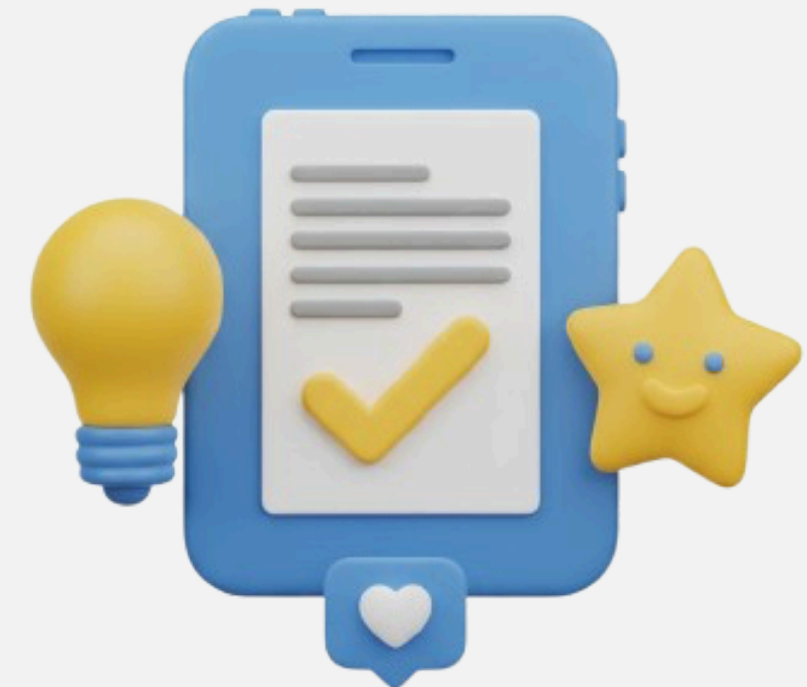


Autumn dissatisfaction

In autumn, satisfaction drops to **66%** vs **72%** in winter (–6 points).
These gaps may help identify dissatisfaction factors specific to this season.



Travelers' satisfaction mainly depends on traveler type, room quality, and seasonal conditions — all key levers for targeted improvement.



Possible improvements



Promote family-friendly hotels

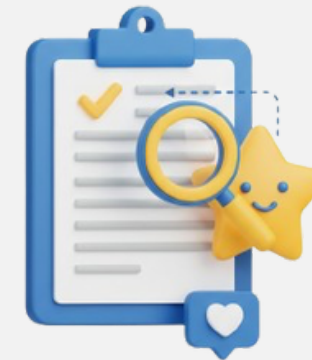
Focus on hotels offering child-oriented amenities (baby beds, kids' menus, family activities).

Families rank among the most satisfied travelers, with satisfaction levels between **66%** and **70%**, slightly higher than the average for other traveler types.



Qualitative studies

Leverage insights from King Rooms (around **75%** satisfied guests) to enhance the quality and guest experience of Single Rooms (55% satisfaction).



Targeted satisfaction survey

Launch a survey focusing on King Rooms (**75%** satisfaction) and Single Rooms (**55%**) to identify the key factors driving these differences



Solo traveler offers

Develop personalized services (simplified check-in, local recommendations, social areas).

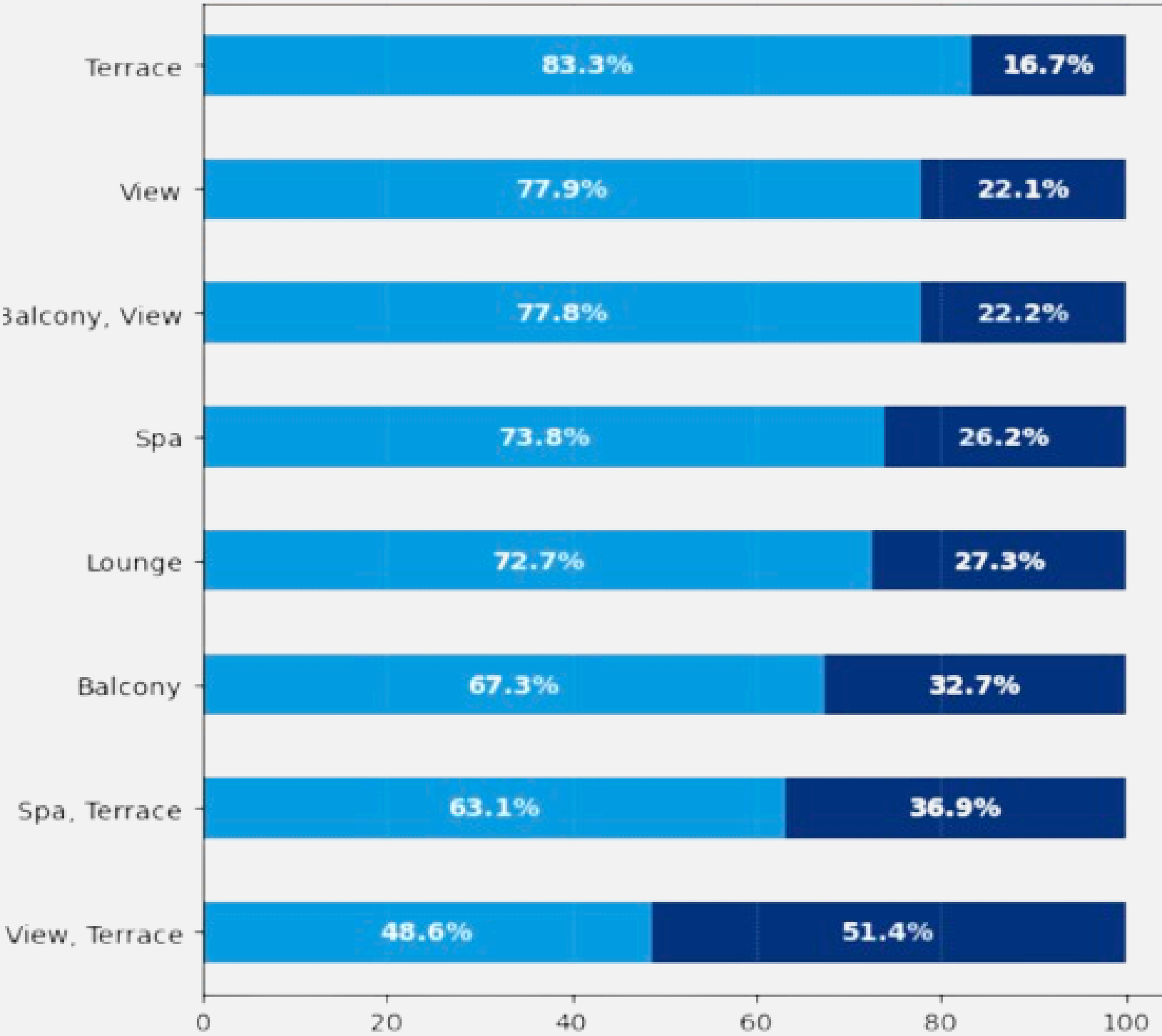
Solo travelers show a satisfaction rate of about **60%**, the lowest among all traveler profiles.



Appendix



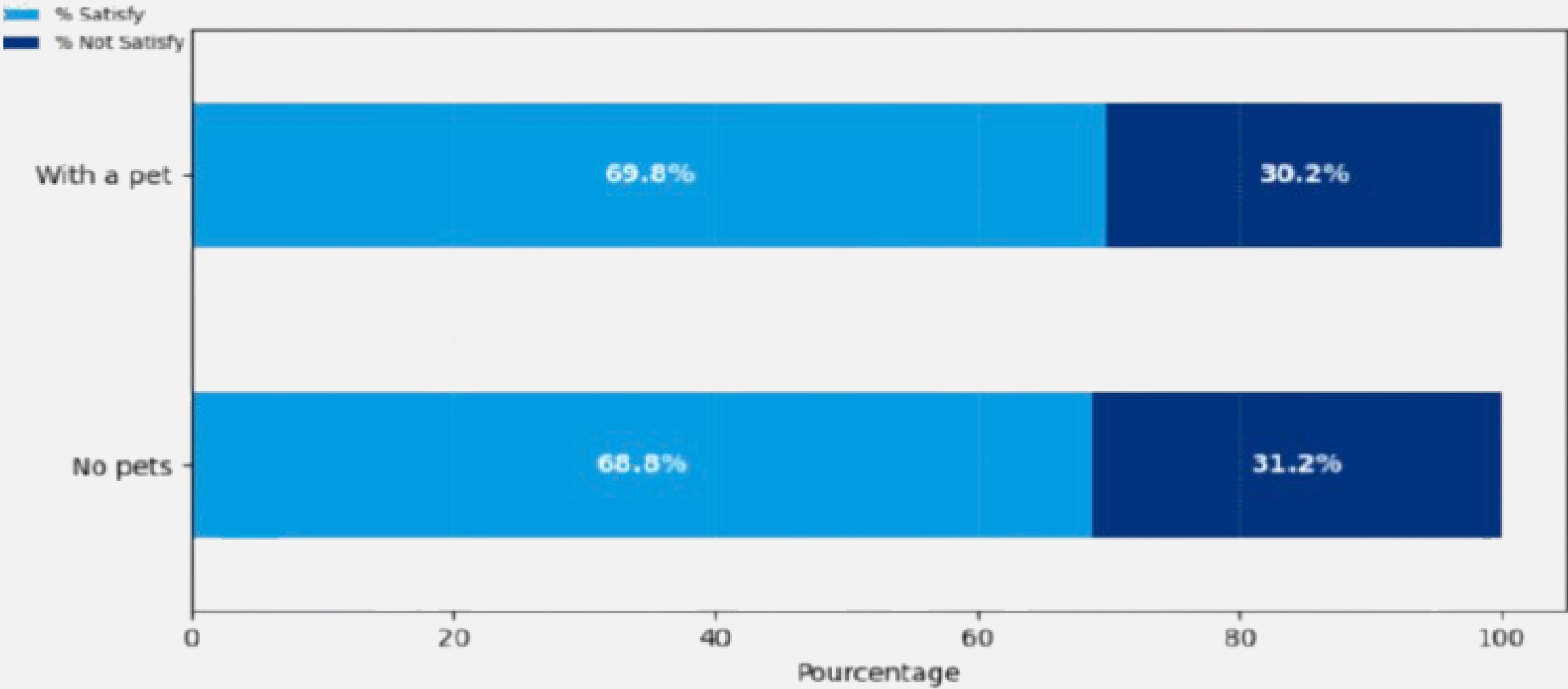
Amenities and satisfaction



The number of reviews including the “features” tag is nearly 21K, representing about 6% of our dataset. Taking this into account, having a terrace appears to be a highly valued feature among travelers, followed by the view, whether or not accompanied by a balcony.

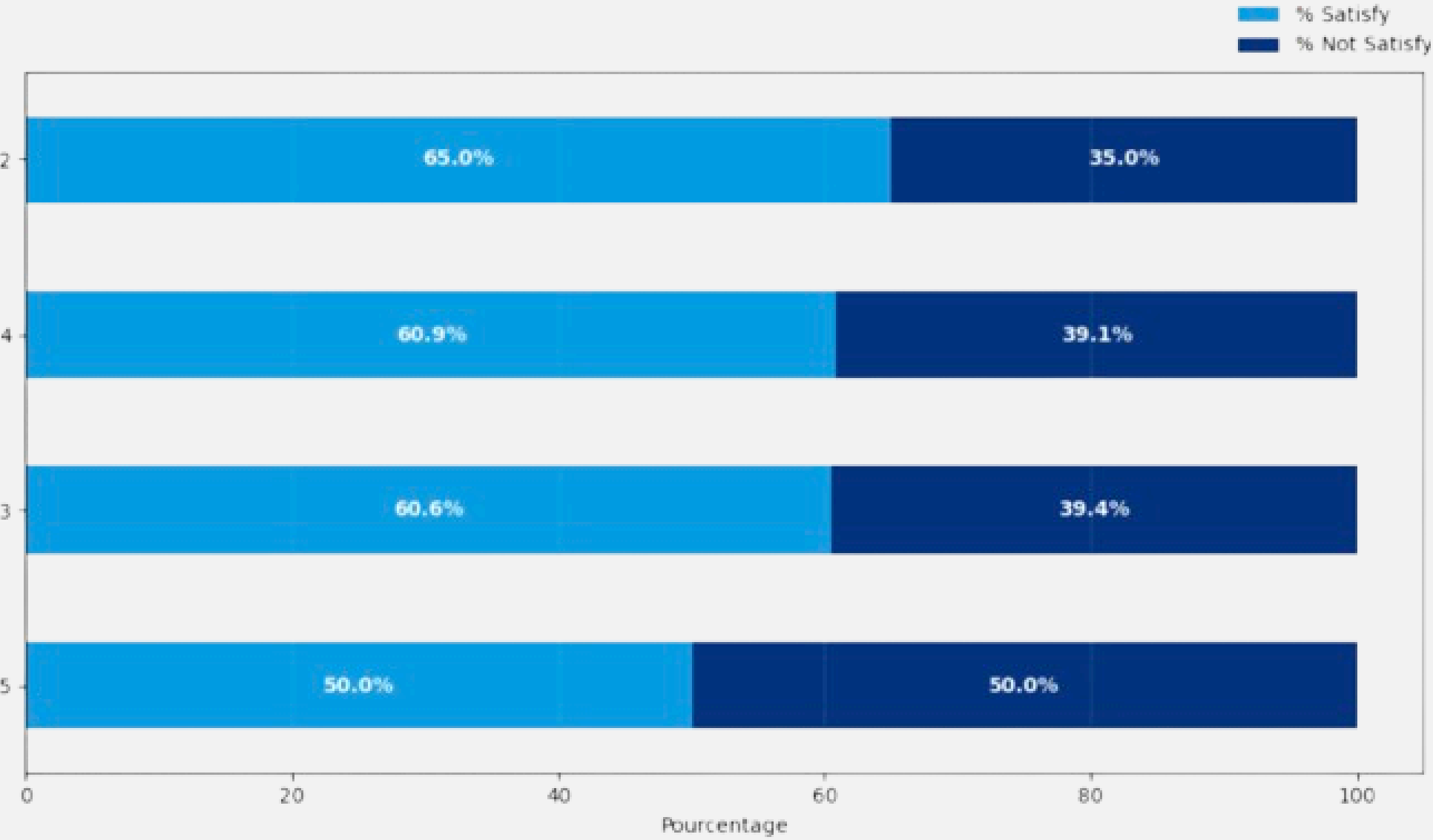


Pets and satisfaction



Pet-friendliness does not appear to impact traveler satisfaction.

Room count and satisfaction



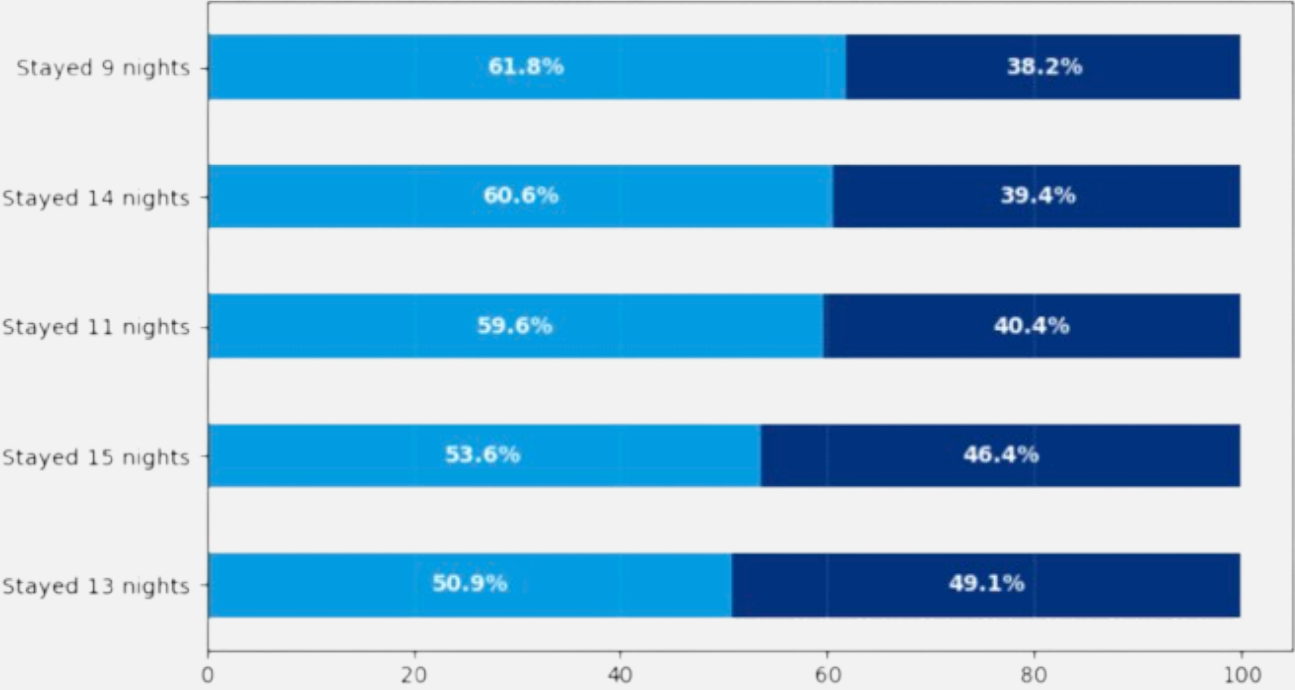
If the sample size had been significant, we could have stated that satisfaction tends to decrease as room size increases.



Stay length and satisfaction

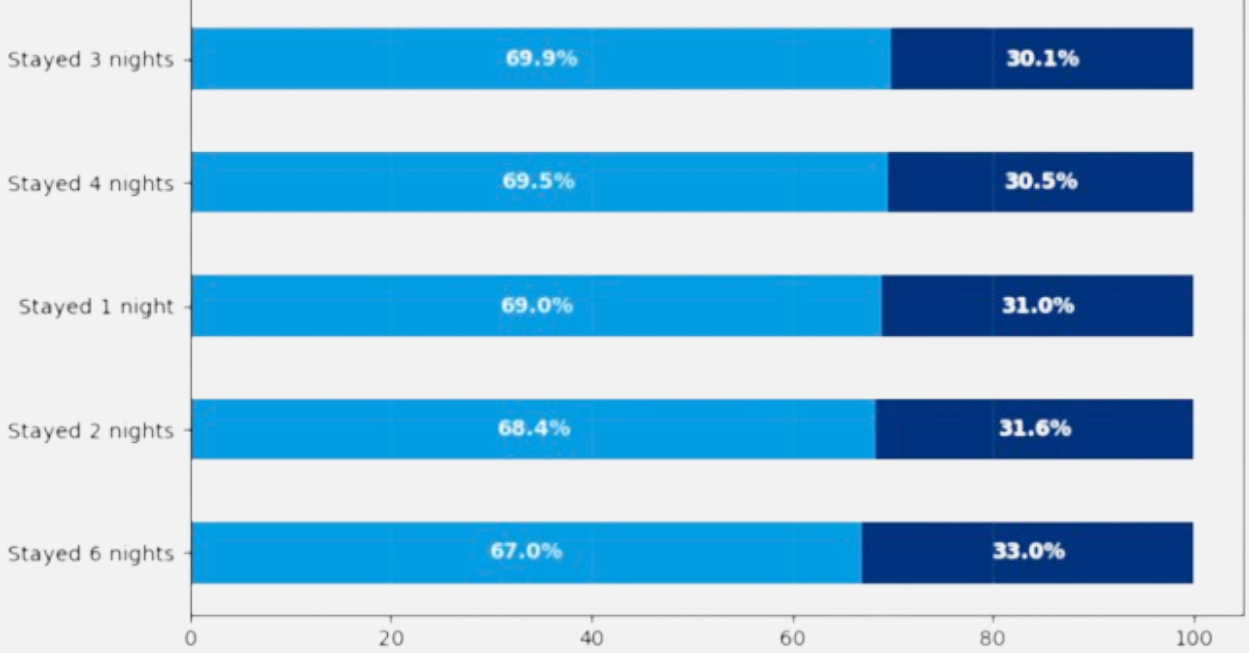


Top 5 stay types with the lowest customer satisfaction



Stays shorter than one week show higher satisfaction levels.

Top 5 stay types with the lowest customer satisfaction



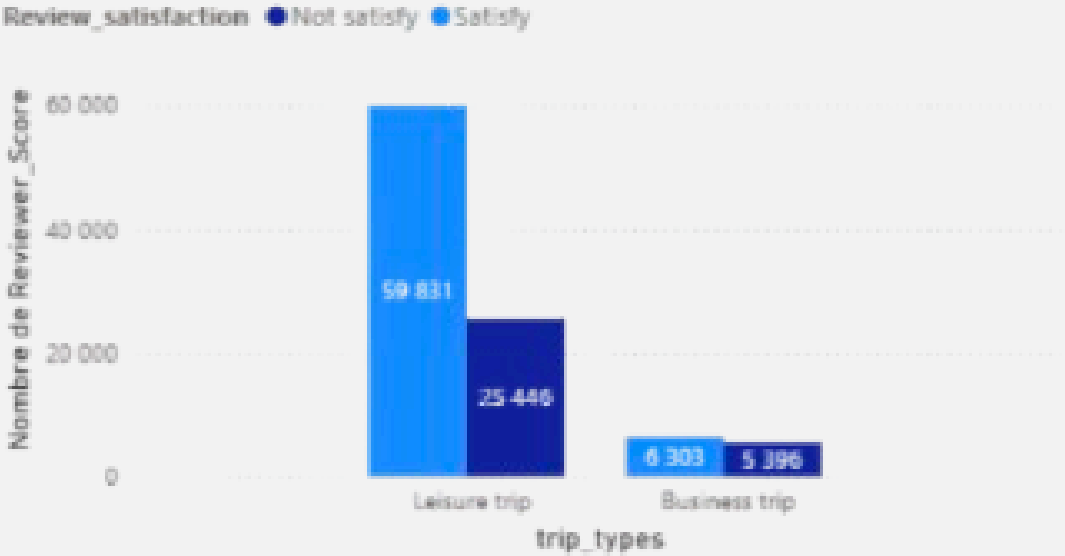
Stays longer than 10 nights show lower satisfaction levels.



Analysis by
trip type



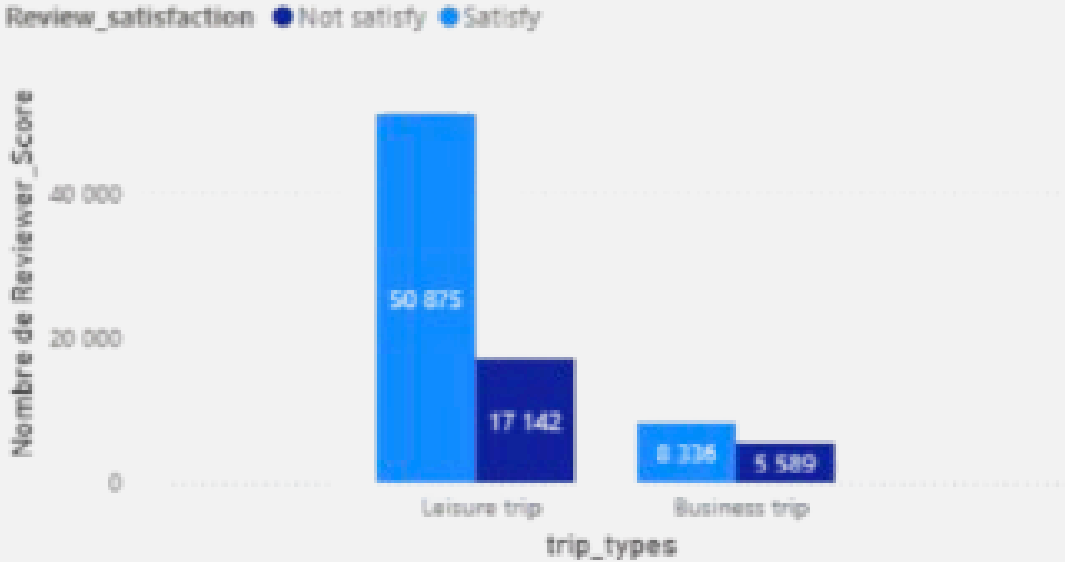
Summer



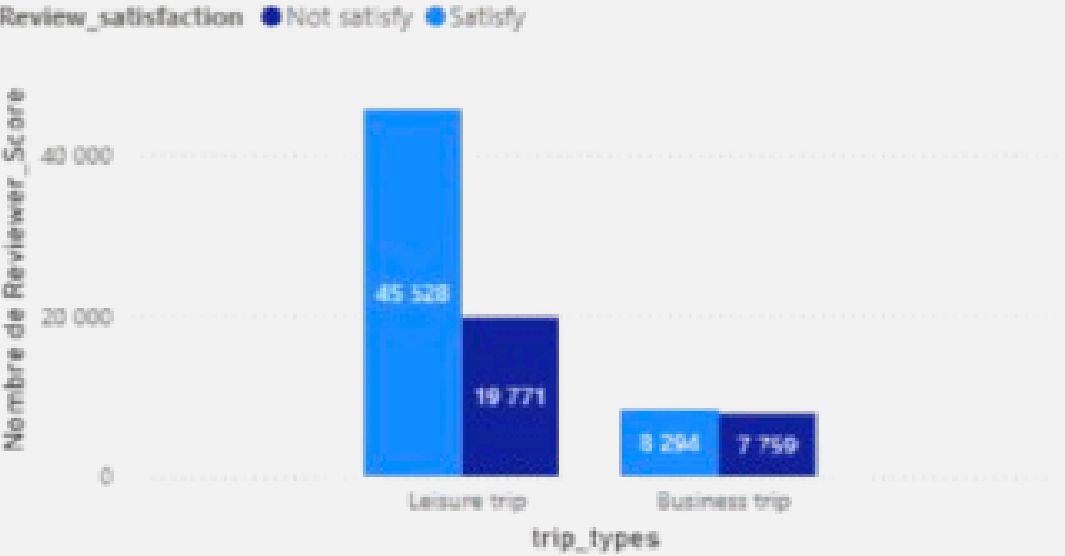
Spring



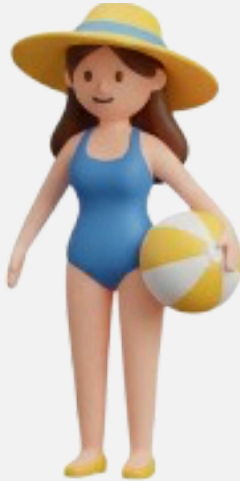
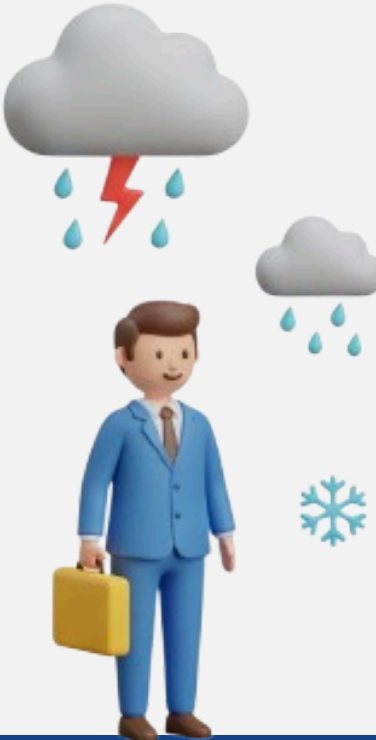
Winter



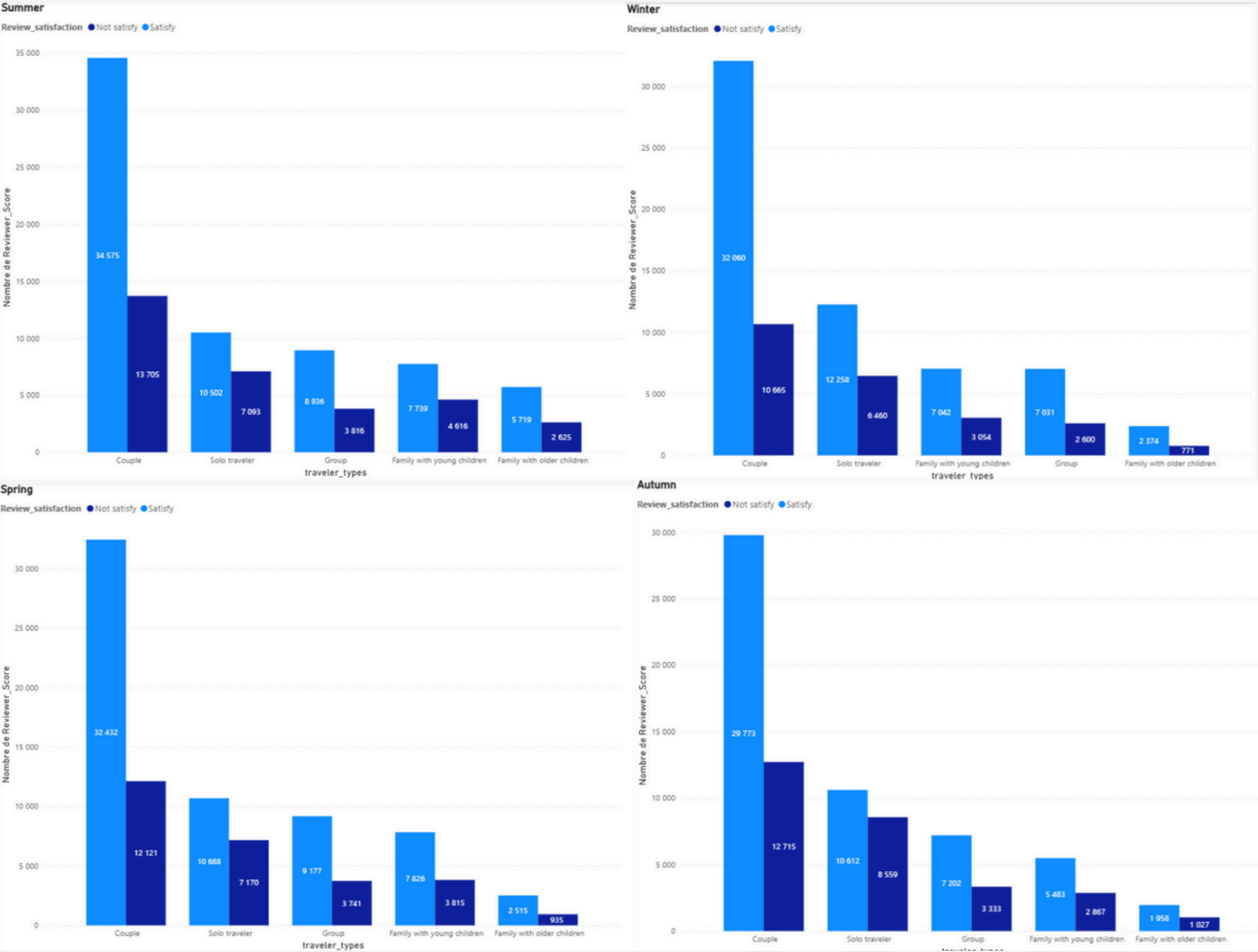
Autumn



Satisfaction mainly comes from leisure trips. However, for this type of travel, autumn remains the season with the lowest overall satisfaction.



Analysis by traveler type



Couples, families, and groups are generally satisfied across all seasons. However, solo travelers show the highest proportion of dissatisfaction, particularly in autumn.

- Reasons for customer dissatisfaction in UK hotels during autumn:
- Unpredictable weather
 - Shorter days
 - Fewer tourist activities
 - Quieter atmosphere
 - Reduced hotel services
 - Perceived lower value for money due to closed activities or disappointing weather

